Solutions to Exercises for Class 7

1a $F(x) = \mathbb{P}(X < x)$. If $F(0) = 0$ then $X \geq 0$ with probability 1, so we can use Markov.

$$\mathbb{P}(X \geq 10) \leq \frac{\mathbb{E}X}{10} \rightsquigarrow \underbrace{\mathbb{P}(X < 10)}_{=0.8} \geq 1 - \frac{\mathbb{E}X}{10} \rightsquigarrow \mathbb{E}X \geq 2.$$

1b Now $\mathbb{E}X = 5$.

$$\mathbb{P}(|X-5| \geq 5) \leq \frac{\mathrm{Var}\,X}{5^2}, \text{ so}$$

$$\mathbb{P}(0 < X < 10) \geq 1 - \frac{\mathrm{Var}\,X}{25}, \text{ so}$$

$$\mathbb{P}(X < 10) \geq 1 - \frac{\mathrm{Var}\,X}{25}$$

$$0.8 \geq 1 - \frac{\mathrm{Var}\,X}{25} \rightsquigarrow \mathrm{Var}\,X \geq 5.$$

2. Let X be the time between two malfunctions:

$X \sim E\left(\frac{1}{30}\right)$. $\quad S_{100} := X_1 + \dots + X_{100}$

By CLT:

$$\mathbb{P}(S_{100} > 3600) = \mathbb{P}\left(\frac{S_{100} - 100 \cdot 30}{\sqrt{100} \cdot \sqrt{30}} > \frac{3600 - 100 \cdot 30}{\sqrt{100} \cdot \sqrt{30}}\right) \approx$$

(by CLT)

$$\approx \mathbb{P}\left(\underbrace{Z}_{\sim N(0,1)} < \frac{-600}{10\sqrt{30}}\right) = \Phi\left(-2\sqrt{30}\right) \approx 0.$$